

Methods: summarized references

Summarized from Zotero Report Methods.htm. Abstracts were replaced with plain-language summaries that emphasize findings, numbers, and conclusions where available. Fields such as Item Type, Library Catalog, URL, Accessed, Extra, ISSN, Date Added, Modified, and Tags were omitted.

1. SPOKEN DIALOGUE TECHNOLOGY: ENABLING THE CONVERSATIONAL USER INTERFACE

First Author: Michael F. McTear

Summary:

Describes the main components of spoken dialogue systems—speech recognition, language understanding, dialogue management, external-source communication, language generation, and speech synthesis—and how they integrate into working systems. The article reviews methods from well-known systems, explores different architectures, specification, design, and evaluation approaches, surveys available development toolkits, and outlines future prospects.

Date: March 1, 2002 Publication: ACM Comput. Surv. Volume: 34 Issue: 1 Pages: 90–169 DOI: 10.1145/505282.505285 <http://doi.org/10.1145/505282.505285>

2. A SURVEY OF AVAILABLE CORPORA FOR BUILDING DATA-DRIVEN DIALOGUE SYSTEMS

First Author: Iulian Vlad Serban

Summary:

Wide survey of publicly available datasets suitable for data-driven dialogue system development. The authors catalogue corpora across domains and discuss how different datasets can train diverse dialogue strategies, noting that transfer learning between datasets and incorporation of external knowledge are promising but underexplored. Evaluation metrics tied to learning objectives are also reviewed.

Date: 2017-03-21 DOI: 10.48550/arXiv.1512.05742 <http://doi.org/10.48550/arXiv.1512.05742>

3. RASA: OPEN SOURCE LANGUAGE UNDERSTANDING AND DIALOGUE MANAGEMENT

First Author: Tom Bocklisch

Summary:

Introduces Rasa NLU and Rasa Core, open-source Python libraries for building conversational software. The tools aim to make machine-learning-based dialogue management and language understanding accessible to non-specialist developers, with design emphasis on ease of use and bootstrapping from minimal or no initial training data.

Date: 2017/12/14

4. FROM ELIZA TO XIAOICE: CHALLENGES AND OPPORTUNITIES WITH SOCIAL CHATBOTS

First Author: Heung-Yeung Shum

Summary:

Traces conversational-system evolution from Eliza (1960s) through task-completion programs and personal assistants to modern social chatbots like XiaoIce. Social chatbots must optimize for both intellectual quotient (IQ) and emotional quotient (EQ), with conversation-turns per session (CPS) proposed as the key success metric. The authors describe technologies from core chat to visual awareness to skills, and discuss dynamic emotion recognition and interpersonal responses as enablers of long, engaging conversations.

Date: 2018-01 Publication: Frontiers of Information Technology & Electronic Engineering
Volume: 19 Issue: 1 Pages: 10-26 DOI: 10.1631/FITEE.1700826
<http://doi.org/10.1631/FITEE.1700826>

5. TECHNICAL METRICS USED TO EVALUATE HEALTH CARE CHATBOTS: SCOPING REVIEW

First Author: Alaa Abd-Alrazaq

Summary:

Scoping review of 65 studies identifying 27 technical (non-clinical) metrics for health chatbot evaluation, grouped by what they assess: the chatbot as a whole (e.g., usability, classifier performance, speed), response generation (e.g., comprehensibility, realism), response understanding (e.g., word error rate), and aesthetics. Survey designs and global usability metrics dominated; the lack of standardization and scarcity of objective measures make cross-study comparison difficult. The authors recommend more frequent use of conversation-log-derived metrics and development of a standardized framework.

Date: 2020-06-05 Publication: Journal of Medical Internet Research Volume: 22 Issue: 6 Pages: e18301 DOI: 10.2196/18301 <http://doi.org/10.2196/18301>

6. TECHNICAL ASPECTS OF DEVELOPING CHATBOTS FOR MEDICAL APPLICATIONS: SCOPING REVIEW

First Author: Zeineb Safi

Summary:

Scoping review of 45 studies on medical chatbot development identifying four main modules: text understanding, dialogue management, database layer, and text generation. The most common technique for text understanding and dialogue management was pattern matching (n=18 and n=25 respectively), and the most common text generation approach was fixed output (n=36) with very few studies generating original output. The review notes a recent shift toward machine-learning-based approaches and calls for research linking clinical outcomes to chatbot development techniques.

Date: 2020-12-18 Publication: Journal of Medical Internet Research Volume: 22 Issue: 12 Pages: e19127 DOI: 10.2196/19127 <http://doi.org/10.2196/19127>

7. EVALUATION FRAMEWORK FOR CONVERSATIONAL AGENTS WITH ARTIFICIAL INTELLIGENCE IN HEALTH INTERVENTIONS: A SYSTEMATIC SCOPING REVIEW

First Author: Hang Ding

Summary:

Systematic scoping review developing an evaluation framework for AI-powered conversational agents used in health interventions. The abstract in the export is truncated; based on the title and publication venue (JAMIA), the work provides a structured framework to guide assessment of conversational AI applied to health contexts.

Date: 2024/02/16 Publication: Journal of the American Medical Informatics Association Volume: 31 Issue: 3 Pages: 746–761 DOI: 10.1093/jamia/ocad222 <http://doi.org/10.1093/jamia/ocad222> PMID: 38070173

8. ADVANCING CLINICAL CHATBOT VALIDATION USING AI-POWERED EVALUATION WITH A NEW 3-BOT EVALUATION SYSTEM: INSTRUMENT VALIDATION STUDY

First Author: Seungheon Choo

Summary:

In test conversations, the patient-education bot scored well for giving correct medical information, clear explanations, and empathy, while the screening bot scored well for communication and exploring emotions. Statistical checks indicated the three-bot evaluator setup could match human reviewers closely enough to use for rapid, low-risk chatbot testing without recruiting real patients.

Date: 2025-02-27 Publication: JMIR Nursing Volume: 8 Issue: 1 Pages: e63058 DOI: 10.2196/63058 <http://doi.org/10.2196/63058>

9. PATIENT-CENTERED AI AGENTS FOR SHARED DECISION-MAKING: A LARGE LANGUAGE MODEL-BASED APPROACH FOR CATARACT MANAGEMENT

First Author: Wenquan Cheng

Summary:

Developed a domain-specific, LLM-powered patient education multi-agent system specifically tailored for cataract management from both open-source and proprietary large language models. System supports personalized, multi-turn dialogue generation, integrates retrieval-augmented generation (RAG) with intention recognition, and employs direct preference optimization (DPO) to align responses with expert-level standards and contextual accuracy. The multi-agent framework showed superior performance compared to traditional LLMs alone: increase in accuracy (12%), reduced rates of hallucination, minimized dissemination of harmful content, improved response comprehensiveness (mean score increase: 0.44, SD 0.10) and understandability (mean increase: 0.19, SD 0.13), and significantly boosted informed decision-making rates to 41.9% from 30.8% with traditional aids ($P = 0.02$).

Date: 2025/06/30 Publication: Investigative Ophthalmology & Visual Science Volume: 66 Issue: 8 Pages: 3945–3945

10. CHATGPT AND LBP – EVALUATING AI-DRIVEN PATIENT EDUCATION IN THE CONTEXT OF INTERVENTIONAL PAIN MEDICINE

First Author: Ahmed Basharat

Summary:

Across 25 standardized LBP questions, clinician reviewers found that "dumbed down" prompts often made ChatGPT's answers less clinically complete, whereas neutral or reference-level prompts preserved accuracy better. Readability scores still landed above typical health-literacy targets, and mistakes were more often missing information than outright false claims. Word count and response length did not reliably signal correctness.

Date: 2025-09-01 Publication: Interventional Pain Medicine Volume: 4 Issue: 3 Pages: 100636
DOI: 10.1016/j.inpm.2025.100636 <http://doi.org/10.1016/j.inpm.2025.100636>

11. STANDARDIZING AND SCAFFOLDING HEALTH CARE AI-CHATBOT EVALUATION: SYSTEMATIC REVIEW

First Author: Yining Hua

Summary:

Systematic review of 266 records producing the Health Care AI Chatbot Evaluation Framework (HAICEF): a hierarchical framework with 3 priority domains (safety, privacy, and fairness; trustworthiness and usefulness; design and operational effectiveness), 18 second-level and 60 third-level constructs covering 271 evaluation questions. Distribution across domains: design/operational effectiveness 40%, trustworthiness/usefulness 39%, safety/privacy/fairness 21%. The framework accommodates both patient-facing and back-office chatbot use cases; planned next steps include prospective validation and Delphi consensus.

Date: 2025-11-07 Publication: JMIR AI Volume: 4 Issue: 1 Pages: e69006 DOI: 10.2196/69006
<http://doi.org/10.2196/69006>

OVERALL SYNTHESIS (FROM SUMMARIES, EMPHASIZING BOLDED CONCLUSIONS)

i) Main findings

Dialogue system foundations and chatbot architecture. The literature traces conversational AI from early spoken dialogue systems built around six core components (speech recognition, language understanding, dialogue management, external-source communication, language generation, speech synthesis) through to modern social chatbots optimizing for both IQ and EQ with conversation-turns per session as a success metric. Medical chatbot development has historically relied on pattern matching for text understanding and dialogue management and fixed output for response generation, though a recent shift toward machine-learning-based approaches is evident. Open-source toolkits such as Rasa have lowered barriers by making ML-based dialogue management accessible to non-specialist developers with minimal initial training data. Publicly available dialogue corpora span many domains, though transfer learning between datasets and incorporation of external knowledge remain underexplored.

Evaluation and metrics. Evaluation of health chatbots is hampered by lack of standardization: across 65 studies, 27 distinct technical metrics were identified, dominated by survey designs and global usability scores with few objective, conversation-log-derived measures. Two frameworks aim to address this: a systematic scoping review in JAMIA proposes a structured evaluation framework for AI-powered conversational agents in health interventions, and the more recent HAICEF framework organizes 271 evaluation questions across 3 priority domains (safety/privacy/fairness, trustworthiness/usefulness, design/operational effectiveness) with 18 second-level and 60 third-level constructs, accommodating both patient-facing and back-office

use cases. An AI-as-evaluator approach (three-bot system) aligned closely enough with human reviewers to support rapid, low-risk chatbot testing without recruiting patients.

Applied health chatbot performance. In a domain-specific, multi-agent LLM system (cataract management) with retrieval-augmented generation and preference alignment, performance exceeded traditional LLM-only baselines on accuracy (+12%), hallucination reduction, comprehensiveness, understandability, and informed decision-making (41.9% vs 30.8%). For ChatGPT-style answers to standardized LBP questions, oversimplified ("dumbed down") prompts reduced clinical completeness whereas neutral or reference-level prompts preserved accuracy better; response length and word count did not reliably signal correctness.

ii) Gaps and opportunities

The literature provides strong architectural and methodological scaffolding for building and evaluating health chatbots, but most work remains outside musculoskeletal and LBP domains. Key gaps include: the absence of standardized, reproducible evaluation metrics across chatbot studies (despite two emerging frameworks, neither has been prospectively validated); the continued dominance of pattern matching and fixed-output generation in medical chatbots, lagging behind advances in data-driven and LLM-based dialogue; and the observation that clinical outcome evidence is rarely linked to chatbot development techniques. Opportunities include: validating HAICEF and similar frameworks in MSK-specific deployments with prospective data and Delphi consensus; adapting multi-agent + retrieval + preference alignment architectures to MSK education, triage, and shared decision-making with patient-centered outcome measures; leveraging open-source toolkits (e.g., Rasa) and available dialogue corpora to bootstrap MSK-specific conversational agents; developing conversation-log-derived metrics as complements to usability surveys; and testing prompt and readability guidelines for LBP/MSK against both clinician and patient endpoints, since readability and length are unreliable proxies for correctness.

###iii) LBP-related datasets

Open-LBP-RF- Clinical note dataset annotated for LBP imaging risk factors
<https://huggingface.co/datasets/Aman-J/OPEN-LBP-RF/viewer/default/train>
https://www.youtube.com/watch?v=T_z2TFuvy4U

Italian LBP Consultation Dataset Each record consists of textual data from three primary clinical sources: radiological reports (MRI and Xray) and consultation notes. MRI reports describe spinal anatomy and pathology; X-ray reports focus on vertebral alignment and bone structure; consultation notes provide narrative summaries written by orthopedic specialists during outpatient visits. Demographic variables, including age and sex, are also available for each patient. https://clic2025.unica.it/wp-content/uploads/2025/09/11_main_long.pdf

PMR-Q&A Q&A dataset of PMR-themed questions derived from scientific texts Could filter selectively for LBP-related questions <https://doi.org/10.3390/bioengineering13010125>

One in a Million Dataset Primary care consultation archive 327 unselected adult patients (18-96 years of age) consulting with 23 GPs in 12 different general practices; includes verbatim transcripts <https://www.bristol.ac.uk/primaryhealthcare/researchthemes/one-in-a-million/>

MedDialog Conversations between doctors and patients English dataset with 0.26 million conversations, 0.51 million utterances, 44.53 million tokens, covering 96 specialties of diseases <https://huggingface.co/datasets/bigbio/meddialog>

i2b2/n2c2 datasets Clinical Semantic Textual Similarity (ClinicalSTS) corpus Medical Concept Normalization (MCN) corpus <https://portal.dbmi.hms.harvard.edu/data-challenges/>

Scraping Reddit or HealthUnlocked <https://www.reddit.com/r/backpain/>
<https://about.healthunlocked.com/>